



WEEK 2: ASSIGNMENT 2

Question 1:

The main drawback of a definite time delay relay is that

- a) it takes a long time to clear close-in faults.
- b) it is unable to clear far-end faults.
- c) both (a) and (b).
- d) None of these

Answer: (a)

Question 2:

The limitations of electromechanical relays include

- a) presence of moving parts leading to friction issues.
- b) low torque production.
- c) high burden and power consumption for auxiliary mechanisms.
- d) All of these

Answer: (d)

Question 3:

A 60 MVA, 33 kV/66 kV, star/delta power transformer is to be protected by differential protection. The continuous current carrying capacity to restraining coils of the differential relay should not exceed 5 A. CT ratio is 3000/5 on the 33 kV side. What would be the current fed into the pilot wires of the CT secondary in the 33 kV side?

- a) 2.13 A
- b) 3.03 A
- c) 4.23 A
- d) 4.85 A

Answer: (b)

Solution:

On 33 kV side,

$$I_F = \frac{60000}{\sqrt{3} \times 33} = 1049.7 \text{ A}$$

CT ratio: 3000/5

$$I_{F(\text{secondary})} = \frac{1049.7}{\left(\frac{3000}{5}\right)} = 1.75 \text{ A}$$

Since, 33 kV is star connected and CT secondary will be connected in Δ . Hence, current fed into pilot wires is:

$$= \sqrt{3} \times 1.75 = 3.03 \text{ A}$$

Question 4:

In continuation of question number 3, what could be the possible CT ratio on 66 kV side?

- a) 200/5
- b) 500/5
- c) 1000/5

d) 1500/5

Answer: (c)

Solution:

On 33 kV side,

$$I_F = \frac{60000}{\sqrt{3} \times 66} = 524.85 \text{ A}$$

Current in secondary of CT's = Current in pilot wires

Since 66 kV side CT secondary are connected in star with current = 3.03 A

$$CT \text{ ratio} = \frac{524.85}{3.03} = 173.22$$

Take CT ratio = 200, Secondary current = 5 A,

Hence, the CT ratio on 66 kV side = 1000/5

Question 5:

Which is the closest relay characteristic to fuse/MCCB?

- a) Normal inverse
- b) Very inverse
- c) Extremely inverse
- d) All of these

Answer: (c)

Question 6:

In an electromagnetic type overcurrent relay having an inverse definite minimum time (IDMT) characteristic, the minimum-time feature is achieved by

- a) saturation of the magnetic circuit.
- b) proper mechanical design.
- c) an appropriate time-delay element.
- d) electromagnetic damping.

Answer: (a)

Question 7:

Fig. 1 shows the single line diagram of a portion of a radial distribution system. The plug setting (PS) of $R_3 = 75\%$ of CT secondary. The TDS of $R_3 = 0.1$. The normal range of PS is 50-200% of 1 A in steps of 25%, whereas the TDS setting range is 0.1 to 1 s in steps of 0.05. What would be the PSs of the relays R_1 and R_2 ?

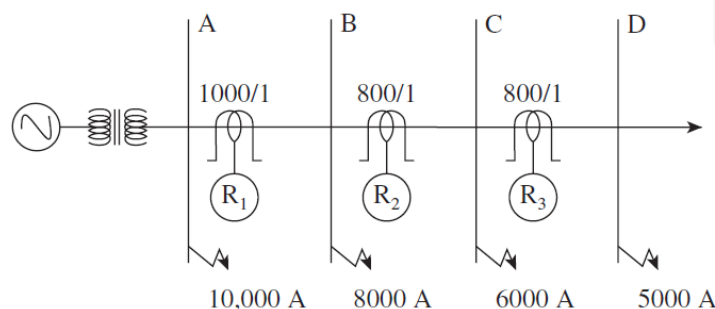


Fig.1 Single line diagram of radial distribution system

- a) 75% and 100% of relay rating, respectively.
- b) 100% of relay rating for both.
- c) 100% and 75% of relay rating, respectively.
- d) 75% of relay rating for both

Answer: (b)

Solution:

Plug setting (PS) of $R_3 = 75\%$

CT ratio (R_1) = 1000/1, CT ratio (R_2) = 800/1, CT ratio (R_3) = 800/1

Pickup current of R_3 in secondary = $0.75 \times 1 = 0.75$ A

Pickup current of R_3 in primary = $0.75 \times 800 = 600$ A

For proper coordination:

R_2 must have a pickup current greater than R_3

R_1 must have a pickup current greater than R_2

If PS of R_2 is set at 75% then the pickup current of R_2 in primary = $0.75 \times 800 = 600$ A

Since it is the same as the pickup current of R_3 , which is not desirable for coordination.

So, at the next higher PS of 100%, the pickup current of R_2 in primary = $1 \times 800 = 800$ A

If PS of R_1 is set at 75% then the pickup current of R_1 in primary = $0.75 \times 1000 = 750$ A

Since it is less than the pickup current of R_2 , which is not desirable for coordination.

So, at the next higher PS of 100%, the pickup current of R_1 in primary = $1 \times 1000 = 1000$ A

Question 8:

In continuation of question number 7, assuming the coordination time interval between two successive relays as 0.3 s, what is the Time Dial Setting (TDS) of the relay R_2 ?

- a) 0.10
- b) 0.15
- c) 0.20
- d) 0.25

Answer: (c)

Solution:

$$MP(R_3) = \frac{\text{Fault current}}{PS \times CT \text{ ratio}} = \frac{5000}{0.75 \times \left(\frac{800}{1}\right)} = 8.33$$

$$t_{op}(R_3) = \frac{0.14}{MP^{0.02} - 1} \times TDS \text{ of } R_3 = \frac{0.14}{8.33^{0.02} - 1} \times 0.1 = 0.323$$
$$t_{op}(R_2) = 0.323 + 0.3 = 0.623$$

For the same fault current,

$$MP(R_2) = \frac{\text{Fault current}}{PS \times CT \text{ ratio}} = \frac{5000}{1 \times \left(\frac{800}{1}\right)} = 6.25$$

$$t_{op}(R_2) = \frac{0.14}{MP^{0.02} - 1} \times TDS \text{ of } R_2$$

$$0.623 = \frac{0.14}{6.25^{0.02} - 1} \times TDS \text{ of } R_2$$
$$TDS \text{ of } R_2 = 0.166$$

Nearest higher setting is 0.2

Therefore, $TDS \text{ of } R_2 = 0.2$

Question 9:

A thermal relay generally has

- a) delayed trip action
- b) instantaneous trip action

- c) both delayed and instantaneous trip action
- d) None of these

Answer: (a)

Question 10:

The transient overreach phenomenon in an overcurrent relay increases when the decay of the DC component of fault current is

- a) Constant
- b) Fast
- c) Slow
- d) None of these

Answer: (c)

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